Facial Landmark Analysis for Detecting Visual Impairment in Mobile LogMAR Test

Maximilian Kapsecker, Elena Mille, Florian Schweizer, Jens Klinker, Joe Yu, Alexander Leube, and Stephan M. Jonas

***Abstract*—Visual impairment is a widespread global health issue that affects millions of people across all ages and backgrounds. Timely intervention is essential for the effective management of eye diseases. Smartphones offer the possibility of continuously recording facial gestures during interaction with the device, whereby changes such as squinting of the eyes could indicate progressive vision loss. In this context, a mobile health application was developed to conduct a digital logMAR test while simultaneously capturing real-time facial features. A total of 37 participants took part in a controlled mobile eye test study. The facial landmarks recorded during the test were analyzed to identify patterns that can distinguish between sequences of letters that were read correctly, partially, or not at all. Specifically, explorative data analysis and receiver operating characteristic curves were employed to determine facial landmarks with high discriminative power in relation to reading ability. The predominant facial regions that showed the most significant change under reduced performance during the vision test were the nose, mouth, and cheeks. Notably, the characteristic maximum squinting of the cheeks stood out with an area under the curve of 0.82. The analysis showed the potential of tracking specific facial features for continuous and unobtrusive vision assessment. It motivates to integrate facial feature analysis into an everyday application such as a web browser and to conduct a study in a non-standardized environment on a larger scale.**

***Index Terms*—Vision impairment, mobile health, digital logmar test, facial landmark analysis.**

I. Introduction

VISION plays a pivotal role in our daily lives, surpassing other senses in importance for decision-making [1], [2]. It is essential for recognizing objects, navigating environments, avoiding collisions, and understanding detailed features like facial expressions and body language [3], [4]. Impairments in vision, especially in high-stakes situations where functional sight is critical, can pose significant problems. For instance, there is evidence that older people with visual impairments are more likely to fall and suffer fractures [5] and that driving a car with a visual impairment is dangerous [6]. According to the World Health Organization, at least 2.2 billion people worldwide had impaired near or distant vision in 2019, with more than 1 billion cases preventable or yet to be treated [7]. Vision impairment represents a significant global health challenge, affecting quality of life, social interactions, and economic productivity [7], [8], [9], [10], [11], [12], [13].

In the digital age, the widespread use of computers and screens has increased the demand for visual skills, possibly also hastening visual deterioration [14]. The increasing number of eye diseases such as myopia, which has reached pandemic proportions in East Asia [15], [16] and poses an economic challenge [13], underlines the need for proactive eye care. Evidence-based guidelines also emphasize the importance of regular eye examinations [17]. While the infrastructure for vision assessments is well-established in high-income countries [18], delays in addressing vision impairment remain prevalent [19]. In many developing regions and rural areas, there is also a significant gap in healthcare infrastructure, particularly in access to ophthalmologists and a clinical environment [18], [20], [21].

Smartphones, now ubiquitous globally, present a unique opportunity to partially bridge the healthcare gap [22], [23]. Equipped with sophisticated sensors and computational capabilities, smartphones are increasingly being used in healthcare applications [24]. Applications like Peek Vision[1] have demonstrated potential to enhance healthcare access by offering services like telemedicine, diagnostic tools, and health monitoring [25]. Additionally, smartphones offer accessibility features, such as increased font sizes and more readable fonts, which

[1]https://peekvision.org Accessed: 03.05.2024

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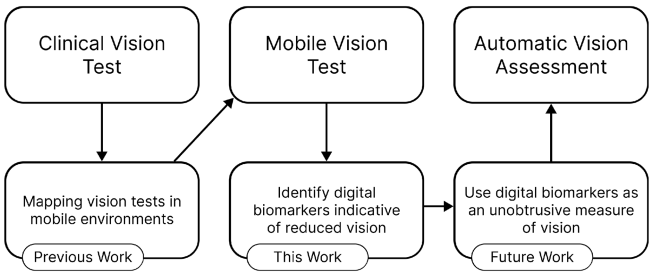


Fig. 1. Overview of how previous research on mobile visual acuity tests connects to the current study and future directions.

can be beneficial for visually impaired individuals [26], [27]. Research has also explored the use of mobile devices for vision assessment. Mobile applications have been developed to conduct visual acuity tests, color vision tests, and contrast sensitivity tests [28], [29], [30], [31], [32], [33], [34]. These applications leverage the device's display to present standardized test charts and collect responses from users, facilitating remote and self-administered vision assessment. *Labiris et al.* have shown through multiple implementations, such as smart tv-based and web-based systems, that digital visual acuity assessments can be used in clinical contexts and as self-assessments at home [35], [36], [37], [38]. Although their implementations have a focus on the greek language, they still highlight the feasibility and correctness of digital visual acuity assessment. Another potential advantage of digital assessments is ease-of-use, especially with non-trained staff [39]. However, *Tofigh et al.* have highlighted potential issues with digital visual acuity assessment implementations, such as calibration of brightness, screen resolution, and color contrast [40].

There is a general consensus in prior research regarding the strong feasibility of conducting standard eye tests in a mobile environment. In particular, *Suo et al. (2022)* provide a systematic review and meta-analysis of mobile versus clinical visual acuity tests, reporting sensitivities of 0.87 (95% CI, 0.79–0.93), 0.86 (95% CI, 0.84–0.87), and 0.85 (95% CI, 0.55–0.98) and specificities of 0.78 (95% CI, 0.70–0.85), 0.91 (95% CI, 0.90–0.91), and 0.98 (95% CI, 0.95–0.99) for the age groups 3–5 years, 6–22 years, and over 55 years, respectively [41]. Building on the findings that demonstrate the feasibility of mobile visual acuity testing, this study investigates potential digital biomarkers that indicate reduced vision. The aim is to provide evidence for the future development of mobile systems for passive vision monitoring (see Fig. 1).

While existing applications enable mobile vision assessments, to the best of the authors' knowledge, there has been limited exploration into the integration of smartphone sensors to detect subtle behavioral changes, such as facial expressions, that may indicate visual impairment. *Chen et al. (2023)* developed a mobile eye examination tool for young children for the early detection of 16 ophthalmic disorders by analyzing gaze direction and facial features in response to visual stimuli [42]. The performance evaluation of the deep-learning based method on an external dataset revealed an AUC of 0.86 and showed robustness while adapting to different environments. As an innovation to current mobile vision assessments, the rationale for recording these indicators stems from the hypothesis that tracking behavioral data could reveal signs of visual impairment and serve as digital biomarkers. Facial expressions, for example, can be key non-verbal indicators of discomfort. Research, such as *Ekman and Friesen's Facial Action Coding System*, has shown that facial expressions like squinting or furrowing of the brow may be associated with emotional experiences [43], which might include difficulty seeing clearly. *Galati et al.* reported an increased frequency of certain facial movements when comparing children with congenital low vision and sighted children [44]. By recognizing these non-verbal cues, it might be possible to infer eyesight issues and initiate further assessments. Additionally, individuals experiencing vision problems may exhibit subtle changes in facial behavior, such as blinking patterns or tilting of the head, which can signal visual discomfort or distress.

Carrying out eye tests with modern mobile devices offers the possibility of simultaneously recording facial landmarks and identifying correlations with visual impairments. These facial landmarks could then assist in detecting visual impairments without explicit eye examinations, by unobtrusively monitoring subtle changes in facial features. Given the frequent use of smartphones in everyday life, this approach is particularly promising as a complement to clinical assessment, which may overlook visual impairments that rather manifest in real-life situations [45]. Regular and unobtrusive tracking of relevant facial features, e.g., while reading on a smartphone, offers a novel method for the potential recognition of vision-related changes, fostering timely interventions and addressing the monitoring of progressive vision loss.

The objective of this study is to design and validate a mobile-based vision assessment application that can accurately detect and quantify vision impairment using standard eye examination techniques while simultaneously recording facial landmarks. Through facial landmark analysis, the application provides behavioral insights into how vision impairment affects facial expressions. Therefore, this study addresses the overarching question: Can the analysis of facial landmarks during a standard vision test performed with mobile devices draw conclusions about readability – possibly related to visual impairment? The long-term aim is to integrate facial feature tracking into everyday technologies, like web browsers and e-readers, for the detection of vision-related behavioral changes, providing timely alerts to users.

To adequately answer this question, a well-designed study needs to be conducted. Studies in an uncontrolled setting can be influenced by numerous confounding factors that can obscure the true impact of visual impairments on facial landmarks. In an uncontrolled environment in which participants use the application alone, different lighting conditions, the angle and distance at which the device is positioned and the subjective assessment of their own eyesight could play a role. In contrast, a standardized environment allows for the controlled assessment of these factors, ensuring that the contributions of facial landmarks are more isolated, pronounced and significant. This controlled approach is an important first step in accurately assessing the potential of facial feature analysis in the detection of visual

impairment. With the findings of this initial study, an extension to unobtrusive tracking in non-standardized environments can be thoroughly planned. This research aims to achieve the following key objectives:

1) **Mobile Application** Design and implement an application for conducting standardized visual acuity tests on mobile devices.
2) **Facial Landmark Feature:** Incorporate a facial landmark feature into the application for recording and evaluating facial landmarks during the vision tests.
3) **Experimentation and Validation:** Conduct a study to verify the application's accuracy in vision testing and to record facial landmarks.
4) **Data Analysis:** Examine changes in facial behavior, like eyelid squinting [46], during vision tests to assess their correlation with test performance.
5) **Impact Assessment and Recommendations:** Assess the application's utility and accuracy and propose future research directions and improvements.

By achieving these objectives, this research aims to contribute to the advancement of mobile-based vision assessment methodologies and pave the way for cost-effective, accessible, and timely detection of vision impairment using mobile devices.

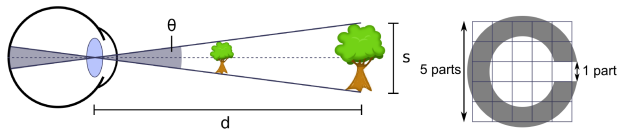
II. Material and Methods

To address the research objectives and answer the proposed questions, the methodology followed three key phases, each described in detail within this section: (1) adapting a standard visual acuity test into a mobile application, incorporating supplementary components such as the recording of facial landmarks and questionnaires; (2) conducting an eye-test study using this mobile application to gather relevant data; and (3) analyzing the collected data, focusing on the relationship between readability and facial gestures, primarily through descriptive and explorative statistical methods.

A. Mobile Application

The application, named *EyeSight*, was specifically developed for the *iPhone 13 Pro* using Swift, the primary programming language for iOS. *EyeSight* was designed to perform standard visual acuity tests while simultaneously capturing facial landmarks to analyze visual behavior. Questionnaires before and after the visual acuity test were used to collect metadata about the participant. The software is available under https://github.com/NumericalMax/viim.

Visual Acuity Test: Visual acuity, the ability of the human eye to recognize fine details, is defined as the resolving power of the eye. It is quantified by the visual angle of the detail, which is the angle formed by the smallest distinguishable two points separated in space of the object seen by the eye (cf. Fig. 2(a)) [47]. For standardized visual acuity tests, a symbol called a Landolt ring is used, which is a C-like optotype where the gap – referred to here as the detail of the letter – is 1/5 of the total height (cf. Fig. 2(b)). In a more clinical set-up of a visual acuity test, letters that fulfill such shape definitions can also be used. For a consistent measurement and for the development of a vision test that is both accurate and adaptable to different display devices such as the iPhone 13 Pro, technical considerations such as screen size and viewing distance lead to limitations and were taken into account in the present study.



(a) Visual Angle. (b) Landolt ring.

Fig. 2. Visual angle and design of a Landolt C optotype illustrating the ratio between detail of the letter (1 part) and total letter size (5 parts).

TABLE I
Letter Sizes for the Visual Acuity Test at Different Levels of Difficulty and a Distance of 3 Meters. Compare [48]

logMAR	Visus	Visual Angle Detail [′]	Visual Angle Letter [′]	Letter Size $[mm]$ for $d = 3m$
1.0	0.1	10.00	50.00	43.63
0.9	0.125	7.94	39.72	34.66
0.8	0.16	6.31	31.55	27.53
0.7	0.2	5.01	25.06	21.87
0.6	0.25	3.98	19.91	17.37
0.5	0.32	3.16	15.81	13.80
0.4	0.4	2.51	12.56	10.96
0.3	0.5	2.00	9.98	8.71
0.2	0.63	1.58	7.92	6.91
0.1	0.8	1.26	6.29	5.49
0.0	1.0	1	5.0	4.36
-0.1	1.25	0.79	3.97	3.46
-0.2	1.6	0.63	3.15	2.75
-0.3	2.0	0.50	2.51	2.19

Visual acuity is often measured scientifically as *logMAR* (logarithmic minimum angle of resolution), which is defined by the equation

$$logMAR = \log_{10}(\theta_d)$$

where θ_d is the visual angle of the detail of an optotype. A lower *logMAR* value signifies better visual acuity. Another metric, Visus V, or decimal visual acuity, inversely represents the visual angle:

$$V = \frac{1}{\theta_d} \Leftrightarrow logMAR = -\log_{10}(V) \tag{1}$$

Human visual acuity is assessed by adjusting the size of details in eye tests in logarithmic steps in accordance with the *DIN EN ISO 8596* standard [48]. Using the conversions from (1), the correct size of the letters s for eye testing can be calculated depending on the visual angle θ and the distance d to the screen. Using trigonometric formulas on the scenario in Fig. 2(a) yields

$$\tan\left(\frac{\theta}{2}\right) = \frac{s}{2 \cdot d}$$

where θ is given in radians and s, d in millimeter (mm). At a distance of 3 m and a *logMAR* stepsize of 0.1, the letter sizes are calculated as given in Table I. With a screen width of 71.5 mm, the *iPhone 13 Pro* can cover all relevant logMAR values at a distance of 3 meters. Considering the limited width,

the maximum distance for performing a vision test on an *iPhone 13 Pro* with a logMAR starting value of 1.0 is 4.92 meters.

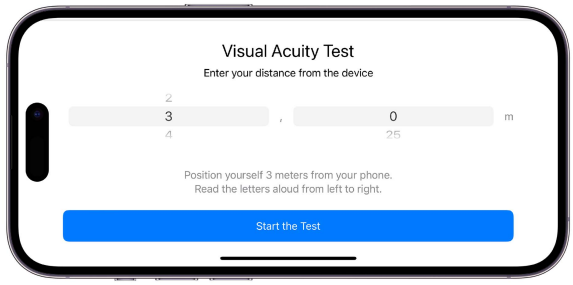
The conversion from letter to font size in points for an *iPhone 13 Pro* required the calculation of a scaling factor. This factor was derived from the definition of points, where 1 m corresponds to approximately 2834.6 points. To better align with actual displayed sizes of the target device, a manual correction factor of 3.45 was determined. The combination of these two factors yielded the overall factor of $9779.37 \frac{points}{meter}$.

A number of five optotypes per row (cf. Fig. 3(c)) are randomly drawn from the following letter pool: A, C, D, E, F, H, L, N, O, P, R, T, V, Y, and Z. The open-source font *Optician Sans* is employed for presenting the optotypes. For a standardized test setting, the brightness of the phone is automatically adjusted to 100% during the eye test.

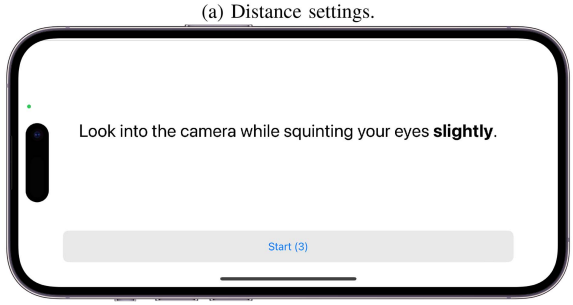
Data collection: During the conduction of the *logMAR* test, the optical sensors, i.e. the device's front-facing camera and the true depth sensor, were used to collect data on a sample basis. Specifically, the face-tracking functionality of Apple's ARKit[2] framework, which interfaces directly with these optical sensors, served as the data delivery layer. Each sample represents a single snapshot of the face that contains metadata such as the timestamp and current *logMAR* level, the tracked face mesh (represented by 1220 three-dimensional vertices), and blendshapes. Latter resembles a preprocessed set of 52 interpreted facial expressions.[3][4] Blendshapes are represented with a title key, e.g. *BrowOuterUpLeft*, and a single floating point number (0–1) to quantify expression strength. These values are sampled at the device's maximum frequency of 60 Hertz.

Squint detection: One of the application's key functional features is the detection of eye squinting. Short-sighted people in particular tend to squint their eyes in order to adjust their vision to see objects in the distance [49]. To reliably detect this, a key functional requirement was the integration of a calibration step to measure individual squinting intensity. During calibration, participants follow a sequence of eye poses, starting from neutral, then progressing to slightly squinting, strongly squinting, and back to slightly squinting (cf. Fig. 3(b)). The application simultaneously records and averages the values of the *EyeSquintRight* and *EyeSquintLeft* facial landmarks provided by ARKit, setting thresholds for each pose. To ensure objectivity, distraction poses such as looking left or right and smiling are included in the calibration process. Calibration relies on a snapshot of each participant's current perception and execution of squinting strength, making exact reproducibility of thresholds unlikely if the experiment is repeated.

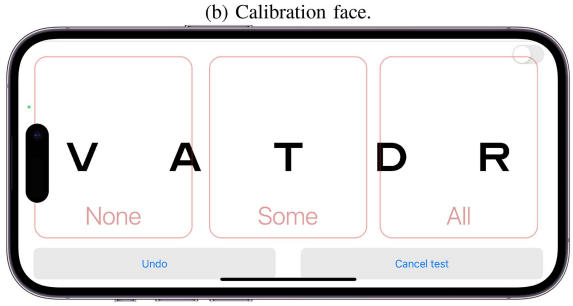
Squint detection is triggered when the participant's thresholds are exceeded. This functionality becomes particularly relevant during the second round of the eye examination cycle, where squinting causes the screen to turn red, preventing the participant from compensating for their vision by squinting – potentially revealing alternative compensatory facial gestures.



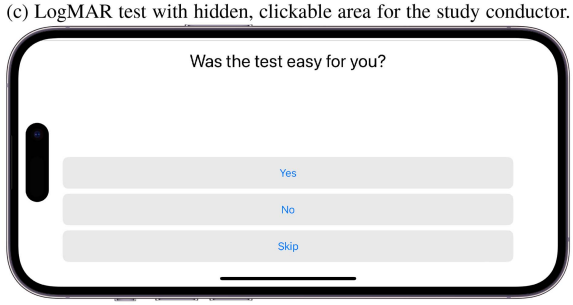
(a) Distance settings.



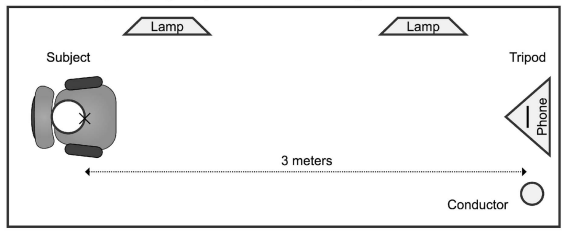
(b) Calibration face.



(c) LogMAR test with hidden, clickable area for the study conductor.



(d) Sample question from the questionnaire.



(e) Study setting.

Fig. 3. Study setting and mobile application.

Questionnaire: Prior to the vision test, a questionnaire collected data on the participant's use of visual aids, including whether they have one, whether they are currently wearing it, and the prescription power in diopters. Following this, three questions explored the participant's subjective assessment of their vision. The participant indicated whether they face challenges

[2]https://developer.apple.com/documentation/arkit. Accessed: 12.12.2024
[3]https://developer.apple.com/documentation/arkit/arfaceanchor/blendshapelocation. Accessed: 25.01.2024
[4]https://arkit-face-blendshapes.com/. Accessed: 07.05.2024

in reading smartphone screens, cinema subtitles or subway signs, and road signs.

In a follow-up questionnaire after the vision test, participants evaluated their performance by indicating whether they found reading the letters challenging overall (cf. Fig. 3(d)) and whether they believe to need a (new) visual aid.

B. Study

The participants were recruited via posters and personal contacts, mainly at the Technical University of Munich (TUM). The study was approved by the TUM ethics committee (2023–509-S-SB). All participants were first informed about the context and procedure of the study, and written informed consent was obtained from all participants beforehand.

For the study the participant was seated at a distance of 3 meters in front of the mobile device (iPhone 13 Pro), which was mounted horizontally – measured with iOS inbuilt spirit level application – on a tripod at the participant's eye level (cf. Fig. 3(e)). The device was adjusted so that it was at the same height as the participant's head and the participant had the feeling that they were looking straight at the screen. For those with visual impairments, the distance from the device was reduced to either two or one meter, depending on the severity of the impairment, to allow the display of higher *logMAR* levels (cf. Fig. 3(a)). To ensure consistent lighting conditions for every test, ceiling lights were switched on. Some participants who wore glasses underwent the procedure twice, once with glasses and once without. The study conductor was located beside the tripod and guided the participant through the test.

After facial calibration and going through the first part of the questionnaire, the actual visual acuity test was carried out twice and with both eyes, i.e., binocularly. Each time the participant was presented with rows of letters that became smaller until the participant could no longer read them. In the second round, any attempts to squint the eyes were automatically detected by the application and prevented by turning the screen red. The participant did not receive any feedback as to whether the letters were recognized correctly. The study conductor secretly labeled the rows of letters by tapping on different areas of the screen, depending on whether *all*, *some*, or *none* of the letters in a row had been recognized (cf. Fig. 3(c)). The second part of the questionnaire concluded the study run. Throughout the study process, the recording of facial landmarks and the calculation of squint thresholds were performed automatically by the system. All other data collection and procedures were manually entered by the study conductor, including completing the questionnaire, labeling whether the letter row was read correctly, initiating the calibration phase when the participant was ready, and navigating to the next screen.

Finally, the study conductor exported the recorded data to the phone's file system and labeled it with a numerical identifier. The raw files were then moved to a computer for further processing.

C. Data Analysis

Preprocessing: The data analysis utilized the recorded samples of the 52 blendshape values, organized into a table and enriched with further metadata such as the participant identifier and logMAR scores. To account for inter-participant variability, such as differences in eye shapes, each blendshape was standardized on a per-participant basis. This standardization enhanced comparability across participants in subsequent analyses.



Letter Row	Raw Values of Sequence for CheekSquintLeft Blendshape	Aggregated Blendshape				Label
		Mean	Std.	Minimum	Maximum	
2	[0.1, 0.04, 0.06, 0.21, 0.13, 0.19, 0.15, ...]	0.13	0.06	0.04	0.21	All
3	[0.09, 0.11, 0.1, 0.13, 0.12, 0.12, 0.14, ...]	0.12	0.02	0.09	0.14	All
4	[0.08, 0.46, 0.62, 0.31, 0.83, 0.52, 0.5, ...]	0.47	0.22	0.08	0.83	Some
...						
9	[0.41, 0.14, 0.89, 0.9, 0.83, 0.47, 0.73, ...]	0.62	0.27	0.14	0.90	None

Fig. 4. Illustration of the data based on the *CheekSquintLeft* blendshape (one of 52). The sample shows its raw data and the four aggregated metrics: mean, standard deviation (std.), minimum, and maximum. The images depict examples of facial snapshots captured while reading the letter rows 2, 3, 4, and 9.

Samples indicating mouth opening, identified by the *JawOpen* blendshape (typically due to speech), were excluded to prevent bias. Talking can influence other facial features and potentially mask the true changes relevant to letter-reading ability during vision tests. While speaking directly relates to reading ability and can bias results, other potential artifacts such as gaze direction and head movement were not systematically excluded. Gaze direction might itself indicate reduced readability, and head movements were minimal since participants generally remained focused on the mobile screen, making it challenging to establish thresholds for excluding such deviations.

The blendshapes were recorded at ARKit's highest resolution of 60 Hz, allowing for detailed capture of facial gestures, even those of short duration. This high-resolution recording provided greater flexibility in data analysis, resulting in multiple snapshots of blendshape values per participant and per letter row. The raw blendshape values were recorded as a sequence and then aggregated by participant and letter row using the mean, standard deviation, minimum, and maximum. In the final dataset, the *Label* column annotated the reading ability for each letter row as *all*, *some*, or *none* of the letters, along with metadata from questionnaires and information on whether squinting was permitted. To minimize biases from initial unfamiliar testing conditions and learning effects, the first presented row of letters shown to each participant was excluded from the dataset.

Fig. 4 exemplifies the process of data recording and feature engineering for a single participant, illustrating how raw blendshape sequences were transformed into aggregated blendshapes for analysis.

Explorative Data Analysis and Statistical analysis: To facilitate a robust exploratory analysis, the blendshapes associated with facial regions – jaw, eye, mouth, cheek, forehead, and nose – were averaged into a single representative value. Blendshapes associated with the tongue were omitted from the analysis based on the assumption that all samples involved closed mouths.

Timeseries plots of the facial regions were created in order to visually examine the change with reduced readability of the letters. Furthermore, boxplots and analysis of variance (ANOVA) were employed to discern any significant differences in facial regions based on the participants' ability to recognize *all*, *some*, or *none* of the letters in a row. The prerequisites for a valid one-way ANOVA – (1) independence of samples, (2) normality of distribution, and (3) homoscedasticity – were met in the dataset. The null hypothesis, asserting no difference in regional scores among the groups, was rejected at a significance level of 5%.

Assessing Sensor Accuracy: Previous work revealed a strongly decreasing reliability of the true depth sensor in dependence of increasing participant-device distance [50]. To verify the accuracy of facial landmark detection for a distance of three meters, four authors performed three predefined gestures with a repetition count of five. The gestures were raising the brows, smiling, and squinting the eyes. Subsequently, the blendshapes, in which a change was anticipated were visually examined regarding signal deflection and the number of repetition counts.

Facial Landmarks as a Predictor of Readability: This study aimed to identify facial landmarks associated with reduced readability, potentially indicative of impaired vision. An interpretable thresholding method was applied to each of the aggregated blendshape values to evaluate their effectiveness in distinguishing between different readability levels. To address the class imbalance – where the *some* ($n = 202$) and *none* ($n = 25$) categories were underrepresented compared to the *all* category ($n = 638$) – the *some* and *none* labels were combined into a single *not all* class, thereby achieving a more balanced sample distribution.

Receiver Operating Characteristic (ROC) curves [51], [52] were employed to assess the ability of blendshape aggregates to differentiate between the *all* and *not all* readability classes. The ROC curve analysis facilitated performance evaluation through the area under the curve (AUC) metric, which quantifies the discriminative power of the blendshape aggregates. AUC values range from 0 to 1, with values further from 0.5 indicating stronger discriminative ability. For example, an AUC of 1.0 for the *mean CheekSquintLeft* blendshape aggregate indicates perfect separation of readability levels.

ROC curves and AUC calculations were performed in two ways:

1) **Individualized:** Each blendshape aggregate (4×52) was evaluated separately for the 45 individual eye examinations. This approach is highly individualized, as each value array (e.g., *CheekSquintLeft Mean* as shown in Fig. 4) was assessed for its ability to separate classes. The resulting AUC values – calculated across 4 aggregations, 52 blendshapes, and 45 eye examinations – were ranked based on performance and visualized using boxplots to summarize the discriminative capabilities of the various blendshapes.
2) **Generalized:** Each blendshape aggregate (4×52) was also evaluated using the combined data from all 45 eye examination tests in one. AUCs and their corresponding 95% confidence intervals (CIs) were calculated using the DeLong test [53] and presented in a table. To evaluate the potential impact of the mild to moderate class imbalance, we repeated the analysis by subsampling the majority class (*all*) to match the sample size of the minority class (*not all*).

A threshold-based classifier was constructed to predict readability based on the blendshape features. To enhance accuracy and reduce feature dimensionality, only blendshape aggregates with AUC values greater than 0.75 were selected for classifier development. Due to the relatively small number of participants, using a traditional hold-out test set comprising approximately 20% of the data was insufficient to achieve statistically significant results. Therefore, performance was assessed using a leave-one-out cross-validation (LOOCV) approach. In the LOOCV method, the classifier was iteratively trained on all participants' eye examination cycles except one, which was used for evaluation. Specifically, the classifier learned thresholds for the selected blendshape aggregates based on the training data from all but one participant's eye examination cycles. These thresholds were then applied to the left-out data, binarizing the aggregated blendshapes as either 0 or 1. To predict the readability of a specific letter row, the binary blendshape values were averaged to obtain a probability score. If the averaged probability exceeded 0.5, the letter row was classified as positive, indicating that it was not readable by the participant. For each left-out participant's eye examination cycle, classifier performance was assessed using balanced accuracy, which is the arithmetic mean of sensitivity and specificity. The resulting balanced accuracies, sensitivities, and specificities were averaged and reported for two scenarios: allowing squinting and not allowing squinting.

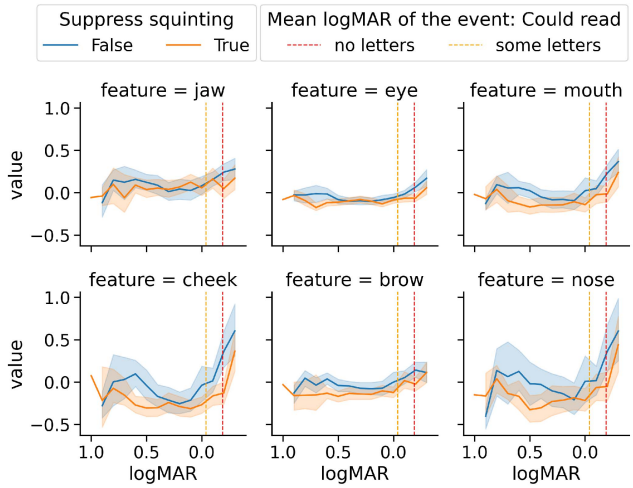
III. Results

A. Study

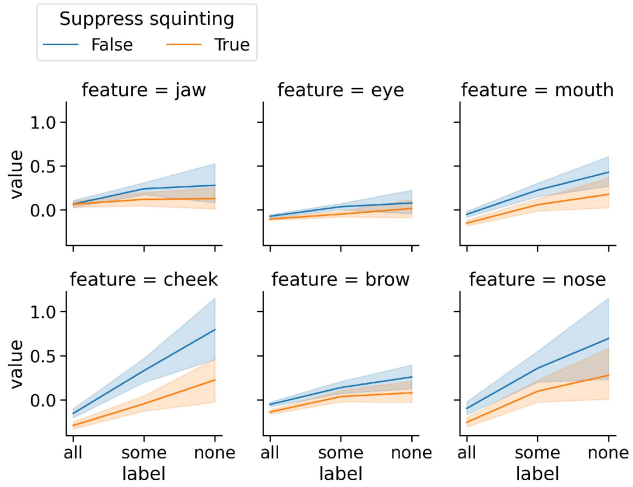
From September 2023 to December 2023, 37 individual participants performed a total of 46 eye tests. Due to metadata corruption, one test was excluded, leaving 45 valid vision tests. The mean diopters (D) of participants was -1.6 D, with a standard deviation of 2.8 D. Of these, 27 tests were conducted with participants who own a visual aid – either glasses or contact lenses – with 16 of the participants using them during the test. Most participants reported that they had no difficulty reading a smartphone screen, which is related to the fact that many either had no visual impairment or were short-sighted. 12 participants stated difficulties when reading cinema titles or recognizing street signs.

The eye tests were primarily conducted at a distance of 3 meters (39 trials), with fewer trials at 2 meters (2 trials), 1 m (2 trials), and 0.5 meters (2 trials). Post-test feedback indicated that 22 participants believed they might need a new visual aid, and 17 found the test challenging.

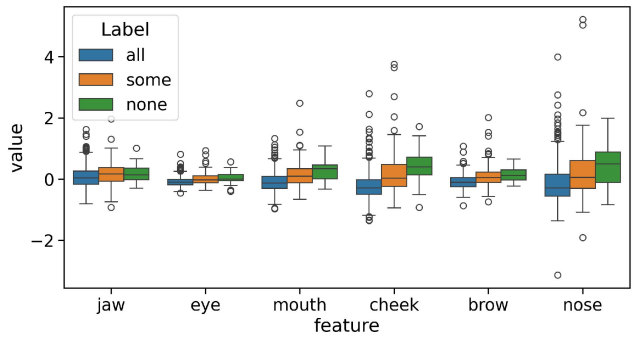
Statistical analysis using Pearson's correlation coefficient revealed a moderate to strong absolute relationship between the *logMAR* values and several binary variables: use of visual aids (-0.32, 95% CI, -0.02 to -0.56), perceived need for new visual aids (0.46, 95% CI, 0.2 to 0.67), ability to read cinema titles (0.49, 95% CI, 0.23 to 0.68), ability to read street signs (0.64, 95% CI, 0.43 to 0.79), and ease of test (0.50, 95% CI, 0.25 to 0.7). Assuming that individuals wearing corrective lenses during



(a) Average facial region change based on logMAR.



(b) Average facial region change based on readability.



(c) Value spread for facial regions in dependence of reading ability.

Fig. 5. Explorative analysis.

the vision test had fully compensated refractive errors (i.e., setting the self-reported diopters to 0.0), the Pearson correlation between *logMAR* values and diopters was -0.80 (95% CI, -0.67 to -0.89).

B. Explorative Data Analysis and Statistics

Fig. 5(a) shows the trend in changes of facial features as a function of logMAR, which represents perceived letter size.

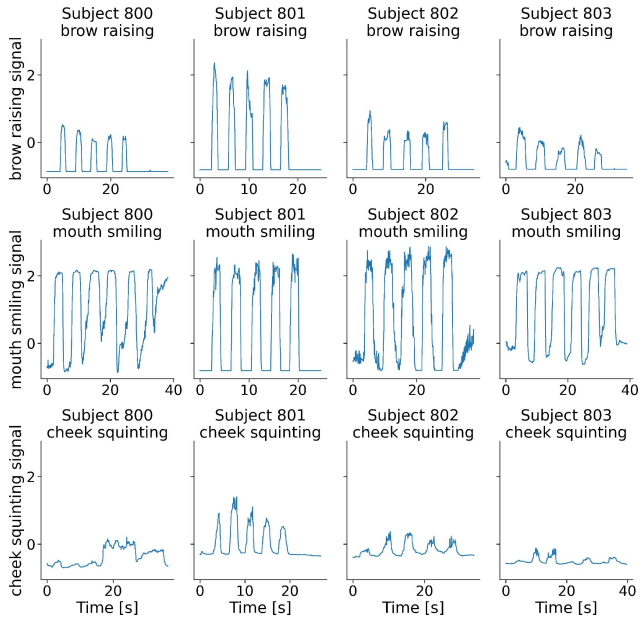


Fig. 6. Signal change when performing predefined actions.

The figure reveals an upward trend, particularly noticeable at logMAR values where participants could read only a few or none of the letters on average. In contrast, Fig. 5(b) presents a similar trend but on a different scale, using the readability class as the x-axis. This approach offers a more individualized perspective, as it accounts for each participant's visual limits, unlike the logMAR-based figure which assumes a uniform standard. This distinction is important because the logMAR figure might yield skewed results if the participants' vision is not homogenous. However, both figures reach a similar conclusion, likely because the participants had relatively similar visual acuity, resulting in a comparable average point where readability declined.

A changing trend of facial landmarks in dependence of letter size and readability aligns with the results of Fig. 5(c). The boxplot reveals a pattern of difference of facial gestures in regards to the three classes that a participant could either read *all*, *some* or *none* of the letters. The visual impression is supported by the results of ANOVA stating that all facial regions have a statistically different population mean regarding the label with all p-values below 0.01.

C. Sensor Accuracy

Fig. 6 illustrates distinct peaks in the measurements for actions associated with eyebrow raising and smiling blendshapes, while the pattern for the cheek squint blendshape during squinting is less pronounced. This subtlety in the squinting data may arise because squinting involves a combination of facial movements and can vary when performed on command by different individuals. However, the graphs clearly showed detectable changes associated with eye squinting. Still, a variation in the deflection of the curves can be observed between the test subjects and imply inter-participant variations.

TABLE II
AUC VALUES AND ASSOCIATED 95% CONFIDENCE INTERVALS OF THE ROC CURVES FOR ASSESSING THE SEPARABILITY OF READABILITY LABELS USING A SINGLE THRESHOLD FOR ALL INDIVIDUALS. BLENDSHAPES AND THEIR AGGREGATES WITH AUC VALUES ABOVE 0.75 WERE COLORED

Aggregation / Blendshape	Squinting allowed				Squinting not allowed			
	Max	Mean	Min	Std	Max	Mean	Min	Std
browDown L	0.7197 (0.66-0.78)	0.5929 (0.53-0.66)	0.6045 (0.54-0.67)	0.737 (0.68-0.79)	0.6667 (0.61-0.73)	0.5219 (0.46-0.58)	0.6183 (0.56-0.68)	0.6885 (0.63-0.75)
browDown R	0.7202 (0.66-0.78)	0.5934 (0.53-0.66)	0.6044 (0.54-0.67)	0.7362 (0.68-0.79)	0.667 (0.61-0.73)	0.5204 (0.46-0.58)	0.6189 (0.56-0.68)	0.689 (0.63-0.75)
browInnerUp	0.7031 (0.64-0.76)	0.6466 (0.58-0.71)	0.5161 (0.45-0.58)	0.6641 (0.6-0.73)	0.7339 (0.68-0.79)	0.6821 (0.62-0.74)	0.524 (0.46-0.59)	0.6845 (0.63-0.74)
browOuterUp L	0.5833 (0.52-0.64)	0.5435 (0.48-0.61)	0.4743 (0.41-0.53)	0.5698 (0.52-0.62)	0.7021 (0.65-0.76)	0.6778 (0.62-0.74)	0.5021 (0.44-0.56)	0.6442 (0.59-0.69)
browOuterUp R	0.5827 (0.52-0.64)	0.5415 (0.48-0.6)	0.4741 (0.41-0.53)	0.5697 (0.52-0.62)	0.7021 (0.65-0.76)	0.6776 (0.62-0.74)	0.5021 (0.44-0.56)	0.6434 (0.59-0.69)
cheekPuff	0.6275 (0.57-0.69)	0.5741 (0.51-0.64)	0.5585 (0.49-0.62)	0.6345 (0.57-0.7)	0.6135 (0.55-0.68)	0.4828 (0.42-0.55)	0.6113 (0.55-0.67)	0.6086 (0.54-0.67)
cheekSquint L	0.8185 (0.77-0.86)	0.7478 (0.7-0.8)	0.547 (0.48-0.61)	0.7735 (0.72-0.82)	0.7819 (0.73-0.83)	0.7228 (0.67-0.78)	0.521 (0.46-0.58)	0.7546 (0.7-0.81)
cheekSquint R	0.8232 (0.78-0.87)	0.7532 (0.7-0.81)	0.547 (0.48-0.61)	0.7875 (0.74-0.84)	0.7883 (0.74-0.84)	0.7249 (0.67-0.78)	0.5126 (0.45-0.58)	0.759 (0.7-0.81)
eyeBlink L	0.5609 (0.5-0.62)	0.5145 (0.45-0.58)	0.5188 (0.46-0.58)	0.532 (0.47-0.59)	0.6229 (0.56-0.68)	0.5375 (0.48-0.6)	0.5149 (0.45-0.58)	0.5751 (0.52-0.63)
eyeBlink R	0.5629 (0.5-0.63)	0.5121 (0.45-0.57)	0.5171 (0.45-0.58)	0.5346 (0.47-0.6)	0.6273 (0.57-0.69)	0.5369 (0.48-0.6)	0.5136 (0.45-0.58)	0.5768 (0.52-0.64)
eyeLookDown L	0.5818 (0.52-0.64)	0.5344 (0.47-0.6)	0.6171 (0.55-0.68)	0.6105 (0.55-0.67)	0.6746 (0.62-0.73)	0.5195 (0.46-0.58)	0.5695 (0.51-0.63)	0.6462 (0.59-0.7)
eyeLookDown R	0.5812 (0.52-0.64)	0.5345 (0.47-0.6)	0.6174 (0.55-0.68)	0.6102 (0.55-0.67)	0.6748 (0.62-0.73)	0.519 (0.46-0.58)	0.5702 (0.51-0.63)	0.6466 (0.59-0.7)
eyeLookIn L	0.5444 (0.48-0.61)	0.4934 (0.43-0.56)	0.4658 (0.4-0.53)	0.5782 (0.51-0.64)	0.5668 (0.5-0.63)	0.5256 (0.46-0.59)	0.5827 (0.52-0.65)	0.5775 (0.51-0.64)
eyeLookIn R	0.5905 (0.52-0.66)	0.5286 (0.46-0.59)	0.5449 (0.48-0.61)	0.5963 (0.53-0.66)	0.67 (0.61-0.73)	0.5656 (0.5-0.63)	0.5386 (0.48-0.6)	0.6326 (0.57-0.69)
eyeLookOut L	0.6103 (0.55-0.67)	0.5673 (0.5-0.63)	0.5134 (0.45-0.58)	0.602 (0.54-0.66)	0.6773 (0.62-0.74)	0.6252 (0.56-0.69)	0.5143 (0.45-0.58)	0.6864 (0.63-0.74)
eyeLookOut R	0.5495 (0.49-0.61)	0.5168 (0.45-0.58)	0.5063 (0.45-0.57)	0.5447 (0.49-0.6)	0.5798 (0.52-0.64)	0.5842 (0.52-0.64)	0.5361 (0.47-0.6)	0.5312 (0.48-0.59)
eyeLookUp L	0.6254 (0.57-0.69)	0.5736 (0.51-0.63)	0.507 (0.45-0.57)	0.5862 (0.54-0.64)	0.5469 (0.48-0.61)	0.5223 (0.46-0.58)	0.5296 (0.47-0.59)	0.5388 (0.49-0.59)
eyeLookUp R	0.6251 (0.57-0.69)	0.5737 (0.51-0.63)	0.5071 (0.45-0.57)	0.5861 (0.54-0.64)	0.5458 (0.48-0.61)	0.5223 (0.46-0.58)	0.53 (0.47-0.59)	0.5388 (0.49-0.59)
eyeSquint L	0.7619 (0.71-0.82)	0.709 (0.65-0.77)	0.5467 (0.48-0.61)	0.7201 (0.66-0.78)	0.6759 (0.61-0.74)	0.5866 (0.52-0.65)	0.5836 (0.52-0.65)	0.6658 (0.6-0.73)
eyeSquint R	0.763 (0.71-0.82)	0.7079 (0.65-0.77)	0.5485 (0.49-0.61)	0.7207 (0.66-0.78)	0.6752 (0.61-0.74)	0.5861 (0.52-0.65)	0.584 (0.52-0.65)	0.6647 (0.6-0.73)
eyeWide L	0.6079 (0.54-0.67)	0.5767 (0.51-0.64)	0.4803 (0.42-0.54)	0.5934 (0.53-0.66)	0.6096 (0.55-0.67)	0.5803 (0.52-0.64)	0.5472 (0.48-0.61)	0.5875 (0.53-0.65)
eyeWide R	0.6075 (0.54-0.67)	0.5767 (0.51-0.64)	0.4804 (0.42-0.54)	0.5925 (0.53-0.66)	0.6077 (0.54-0.67)	0.5787 (0.52-0.64)	0.5472 (0.48-0.61)	0.5865 (0.52-0.65)
jawForward	0.7286 (0.67-0.78)	0.6788 (0.62-0.74)	0.5098 (0.44-0.58)	0.6837 (0.62-0.75)	0.7116 (0.66-0.77)	0.5922 (0.53-0.65)	0.5685 (0.5-0.63)	0.6779 (0.62-0.74)
jawLeft	0.702 (0.64-0.76)	0.5846 (0.52-0.65)	0.5961 (0.54-0.66)	0.6964 (0.64-0.75)	0.6867 (0.63-0.74)	0.5282 (0.46-0.59)	0.6688 (0.61-0.73)	0.7231 (0.67-0.78)
jawOpen	0.5039 (0.44-0.57)	0.5657 (0.51-0.62)	0.5203 (0.46-0.58)	0.5166 (0.45-0.58)	0.5146 (0.45-0.58)	0.5323 (0.47-0.59)	0.5581 (0.5-0.62)	0.5431 (0.48-0.61)
jawRight	0.5342 (0.47-0.6)	0.5188 (0.46-0.58)	0.5048 (0.44-0.57)	0.519 (0.47-0.57)	0.6055 (0.54-0.67)	0.5297 (0.47-0.59)	0.5571 (0.49-0.62)	0.5891 (0.54-0.64)
mouthClose	0.6269 (0.57-0.69)	0.5312 (0.47-0.6)	0.6094 (0.55-0.67)	0.5877 (0.53-0.65)	0.6159 (0.55-0.68)	0.5067 (0.45-0.57)	0.6184 (0.56-0.68)	0.6112 (0.55-0.67)
mouthDimple L	0.7601 (0.71-0.81)	0.7086 (0.65-0.76)	0.5211 (0.46-0.59)	0.6994 (0.64-0.76)	0.7557 (0.71-0.81)	0.6799 (0.62-0.74)	0.492 (0.43-0.56)	0.7051 (0.65-0.76)
mouthDimple R	0.7639 (0.71-0.81)	0.7094 (0.65-0.77)	0.5288 (0.46-0.59)	0.701 (0.64-0.76)	0.7584 (0.71-0.81)	0.6787 (0.62-0.74)	0.4931 (0.43-0.56)	0.7066 (0.65-0.77)
mouthFrown L	0.549 (0.49-0.61)	0.5785 (0.52-0.64)	0.5368 (0.47-0.6)	0.515 (0.46-0.57)	0.5818 (0.52-0.65)	0.5838 (0.52-0.65)	0.5136 (0.45-0.58)	0.5509 (0.49-0.61)
mouthFrown R	0.5012 (0.44-0.57)	0.5322 (0.47-0.59)	0.5313 (0.47-0.59)	0.5163 (0.46-0.58)	0.4645 (0.4-0.53)	0.5539 (0.49-0.62)	0.5137 (0.45-0.58)	0.4845 (0.42-0.55)
mouthFunnel	0.5548 (0.49-0.62)	0.5324 (0.47-0.59)	0.5432 (0.48-0.61)	0.5105 (0.45-0.58)	0.5808 (0.52-0.64)	0.5721 (0.51-0.63)	0.5454 (0.48-0.61)	0.5474 (0.49-0.61)
mouthLeft	0.6288 (0.57-0.69)	0.5324 (0.47-0.6)	0.5483 (0.49-0.61)	0.6264 (0.57-0.69)	0.6762 (0.62-0.74)	0.5625 (0.5-0.63)	0.5647 (0.5-0.63)	0.6593 (0.6-0.72)
mouthLowerDown L	0.7227 (0.67-0.78)	0.6647 (0.61-0.72)	0.5387 (0.47-0.6)	0.7291 (0.67-0.79)	0.738 (0.69-0.79)	0.6685 (0.61-0.73)	0.5145 (0.45-0.58)	0.7195 (0.67-0.77)
mouthLowerDown R	0.7251 (0.67-0.78)	0.663 (0.61-0.72)	0.5459 (0.48-0.61)	0.7278 (0.67-0.78)	0.7305 (0.68-0.78)	0.659 (0.6-0.72)	0.4761 (0.41-0.54)	0.7139 (0.66-0.77)
mouthPress L	0.6769 (0.62-0.73)	0.6614 (0.6-0.72)	0.5026 (0.44-0.57)	0.659 (0.6-0.72)	0.6695 (0.61-0.73)	0.6387 (0.58-0.7)	0.4673 (0.4-0.53)	0.6471 (0.59-0.71)
mouthPress R	0.6786 (0.62-0.74)	0.6554 (0.6-0.71)	0.5092 (0.45-0.57)	0.6583 (0.6-0.72)	0.6682 (0.61-0.73)	0.6352 (0.58-0.69)	0.5283 (0.46-0.59)	0.6474 (0.59-0.71)
mouthPucker	0.5665 (0.5-0.63)	0.5154 (0.45-0.58)	0.5909 (0.53-0.65)	0.5679 (0.51-0.63)	0.5828 (0.52-0.65)	0.524 (0.46-0.59)	0.5937 (0.53-0.65)	0.575 (0.51-0.64)
mouthRight	0.6066 (0.54-0.67)	0.503 (0.44-0.57)	0.5269 (0.46-0.59)	0.6033 (0.54-0.67)	0.5935 (0.53-0.66)	0.5196 (0.46-0.58)	0.5573 (0.5-0.62)	0.5881 (0.52-0.65)
mouthRollLower	0.7521 (0.7-0.81)	0.6987 (0.64-0.76)	0.513 (0.45-0.58)	0.7224 (0.67-0.78)	0.7418 (0.69-0.8)	0.6573 (0.6-0.72)	0.4807 (0.42-0.54)	0.7066 (0.65-0.77)
mouthRollUpper	0.6254 (0.56-0.69)	0.584 (0.52-0.64)	0.6676 (0.61-0.72)	0.6516 (0.59-0.71)	0.5097 (0.45-0.57)	0.6293 (0.57-0.69)	0.6763 (0.62-0.73)	0.5819 (0.52-0.64)
mouthShrugLower	0.7655 (0.71-0.82)	0.7288 (0.67-0.78)	0.5559 (0.49-0.62)	0.7239 (0.67-0.78)	0.716 (0.66-0.77)	0.6946 (0.64-0.75)	0.5158 (0.45-0.58)	0.6739 (0.61-0.74)
mouthShrugUpper	0.7685 (0.72-0.82)	0.741 (0.69-0.79)	0.5375 (0.47-0.6)	0.7326 (0.68-0.79)	0.7325 (0.68-0.79)	0.7097 (0.65-0.77)	0.5145 (0.45-0.58)	0.6885 (0.63-0.75)
mouthSmile L	0.7883 (0.74-0.84)	0.7186 (0.66-0.78)	0.5154 (0.45-0.58)	0.7326 (0.68-0.79)	0.7887 (0.74-0.84)	0.708 (0.65-0.76)	0.5098 (0.45-0.57)	0.7582 (0.71-0.81)
mouthSmile R	0.7902 (0.74-0.84)	0.7225 (0.67-0.78)	0.5201 (0.45-0.59)	0.7399 (0.69-0.79)	0.7914 (0.74-0.84)	0.7122 (0.66-0.77)	0.502 (0.44-0.57)	0.7538 (0.7-0.81)
mouthStretch L	0.6648 (0.61-0.72)	0.6694 (0.61-0.72)	0.5322 (0.47-0.6)	0.592 (0.53-0.66)	0.6828 (0.63-0.74)	0.658 (0.6-0.72)	0.5017 (0.44-0.56)	0.6211 (0.56-0.68)
mouthStretch R	0.6614 (0.6-0.72)	0.668 (0.61-0.72)	0.5352 (0.47-0.6)	0.5982 (0.53-0.66)	0.6762 (0.62-0.73)	0.6541 (0.6-0.71)	0.5074 (0.44-0.57)	0.6179 (0.56-0.68)
mouthUpperUp L	0.7526 (0.7-0.8)	0.7072 (0.65-0.76)	0.5209 (0.46-0.59)	0.7467 (0.69-0.8)	0.7455 (0.69-0.8)	0.7101 (0.65-0.77)	0.5176 (0.46-0.58)	0.7162 (0.66-0.77)
mouthUpperUp R	0.7544 (0.7-0.81)	0.7062 (0.65-0.76)	0.5255 (0.46-0.59)	0.7452 (0.69-0.8)	0.7509 (0.7-0.8)	0.7115 (0.66-0.77)	0.521 (0.46-0.58)	0.7201 (0.66-0.78)
noseSneer L	0.7706 (0.72-0.82)	0.6864 (0.63-0.74)	0.5107 (0.45-0.58)	0.7511 (0.7-0.8)	0.7711 (0.72-0.82)	0.69 (0.63-0.75)	0.4936 (0.43-0.56)	0.7409 (0.69-0.8)
noseSneer R	0.7678 (0.72-0.82)	0.6817 (0.62-0.74)	0.5231 (0.46-0.59)	0.756 (0.7-0.81)	0.7649 (0.71-0.82)	0.6895 (0.63-0.75)	0.4996 (0.44-0.56)	0.7385 (0.68-0.8)
tongueOut	0.566 (0.51-0.62)	0.5443 (0.49-0.6)	0.5298 (0.48-0.58)	0.5172 (0.48-0.55)	0.5448 (0.49-0.6)	0.5373 (0.49-0.59)	0.5214 (0.47-0.57)	0.5185 (0.48-0.56)

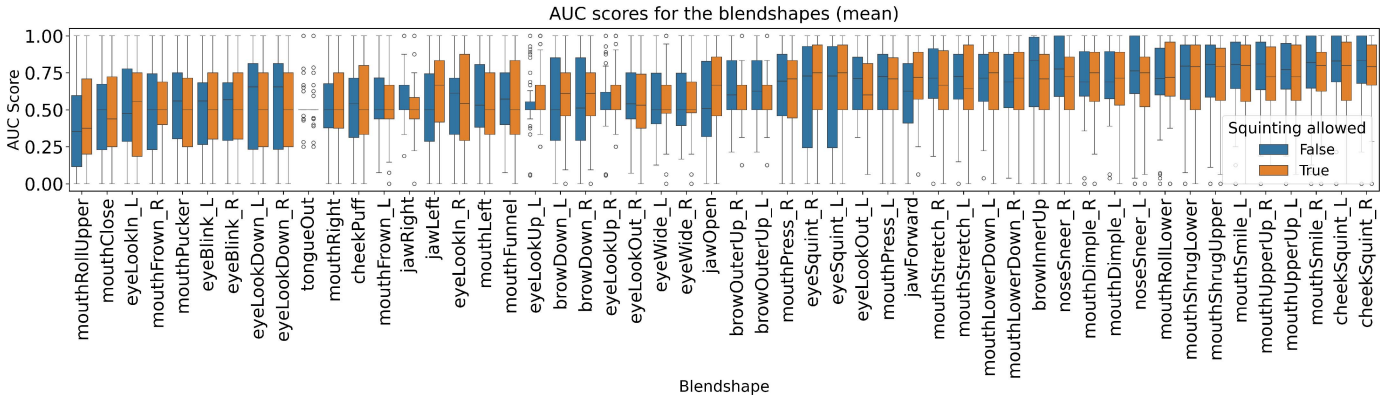
D. Prediction of Readability

Fig. 7 visualizes the AUC score distribution across participants for the aggregated blendshape values. The results indicate that the mean, standard deviation, and maximum aggregations generally revealed blendshapes with substantial class separability on an individual basis. Notably, the maximum *CheekSquintRight* blendshape achieved the highest class discrimination when squinting was permitted, with an average AUC of 0.83 ± 0.17, closely followed by the maximum *CheekSquintLeft* blendshape with an average AUC of 0.82 ± 0.19. The maximum *MouthSmile* (0.77 ± 0.19 for both left and right sides), as well as the standard deviation of *CheekSquintLeft* (0.76 ± 0.24) and *CheekSquintRight* (0.78 ± 0.24), also demonstrated decent performance. When squinting was not allowed, the maximum *MouthSmile* blendshape achieved an average AUC of 0.84 ± 0.22 for the left side and 0.83 ± 0.22 for the right side. Overall, *CheekSquint* blendshapes emerged as the predominant discriminators among all aggregation methods.

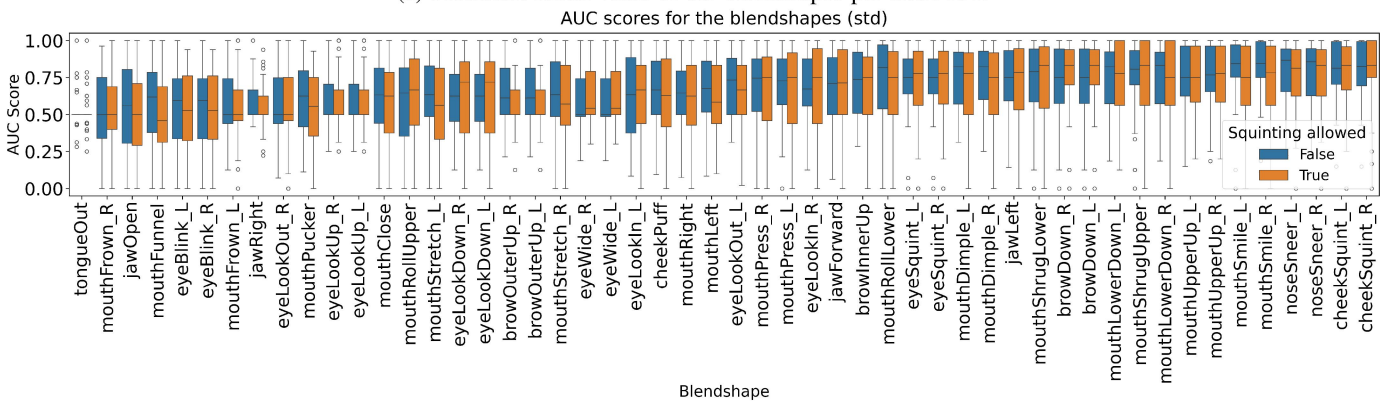
Table II presents the ROC curves and corresponding AUC values calculated for each blendshape aggregate using the combined data from all participants. Blendshape aggregates with AUC values exceeding 0.75 are highlighted, indicating their strong performance in distinguishing between the *all* and *not all* readability classes. The top-performing blendshapes primarily involve maximum deflection during the reading of a letter row, including cheek squint, eye squint, mouth dimple, lower mouth roll, mouth shrug, mouth smile, mouth upper up, and nose sneer. The estimated 95% CIs for the AUCs are approximately ± 0.05, reinforcing the statistical significance of these findings. The repeated analysis with balanced samples revealed slight variations but remained largely consistent with the original results.

Utilizing these high-performing blendshape aggregates in the classification model, LOOCV approach was employed to evaluate performance. The classifier achieved an average balanced accuracy of: (1) 0.7749 (Sensitivity: 0.7879, Specificity: 0.762) when squinting was allowed; (2) 0.7787 (Sensitivity: 0.7378, Specificity: 0.8195) when squinting was not allowed.

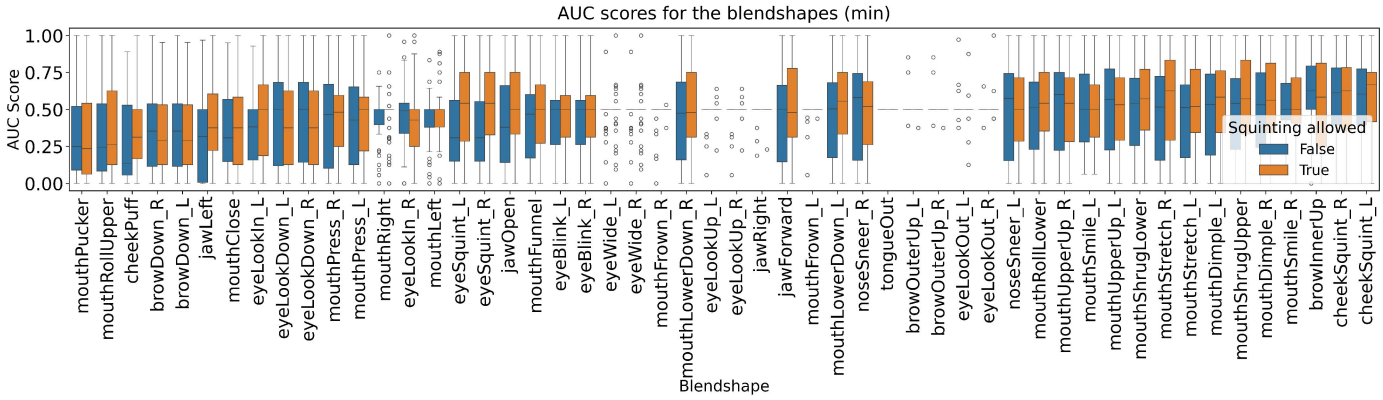
The analysis of blendshape metrics with strong discriminative power revealed several key findings: (1) The mean blendshape aggregates for *CheekSquint*, *MouthUpperUp*, and *MouthSmile* showed the highest average AUC values, indicating that these facial landmarks significantly differed between the *all* and *not all* readability classes when assessed on an individual basis. However, mean blendshape aggregates were not significant in the generalized approach; (2) The standard deviation blendshape aggregates for *CheekSquint* and *NoseSneer* demonstrated high average AUC values in both generalized and individual assessments, indicating that the variation in movement for these landmarks was most distinct between the classes; (3) The minimum blendshape aggregates did not reveal any meaningful patterns, which is expected, as the minimum often represents an inactive facial landmark and thus reflects normal facial behavior; (4) The



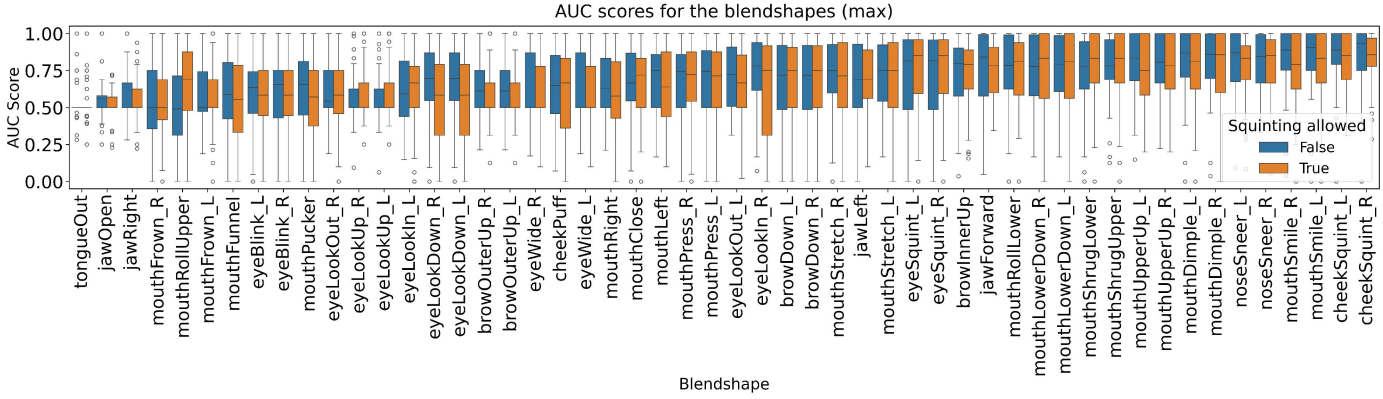
(a) Predictor: Mean value of the blendshapes per letter row.



(b) Predictor: Standard deviation value of the blendshapes per letter row.



(c) Predictor: Minimum value of the blendshapes per letter row.



(d) Predictor: Maximum value of the blendshapes per letter row.

Fig. 7. Distribution of individualized AUC scores of the aggregated blendshapes for readability.

maximum blendshape aggregates for *CheekSquint*, *NoseSneer*, and *MouthSmile* showed the highest AUC values, suggesting that significant deflection of these landmarks was strongly associated with readability levels; (5) A high degree of consistency was observed between the left and right blendshape data points when both were available. This alignment is attributed to the natural symmetry of facial movements, reinforcing the robustness and reliability of the results.

IV. Discussion

A. Feasibility

The results of this study demonstrate the feasibility of adapting standard eye tests for mobile environments. The letter sizes were calibrated to adhere to medical standards, while the experimental setting simulated an optometrist's environment. The detection of squinting was effective for most participants and the overall signal accuracy was sufficient to allow analysis from a distance of at least three meters (cf. Fig. 6). Although the test could not detect fine-grained differences, the participants' *logMAR* values converted to visual acuity (cf. Table I) correlated well with their self-reported diopters. This aligns with previous findings on the performance of mobile visual acuity tests [41] and highlights the significant potential of mobile applications for assessing myopia and other visual impairments.

In addition, the pilot study provided promising results in detecting behavioral changes related to letter perception and reading accuracy. Visual inspection of facial regions, particularly those encompassing individual facial features, indicated significant changes when participants struggled with letter sizes, typically starting from a *logMAR* of 0.0. The average *logMAR* of 0.0 generally stands for a good visual acuity of the participants, which is not surprising due to the predominance of young participants [54]. The consistently low orange line in 5(a) and 5(b) is due to squinting being prohibited, resulting in less curve deflection compared to when participants were free to use their facial expressions.

The average AUC scores further confirmed the presence of discriminative factors related to letter recognition and readability. Cheek squinting emerged as the predominant blendshape for distinguishing between classes. This finding aligns with early work by *Bradburne (1909)*, who associated squinting with refractive errors, fusion defects, or other vision impairments [49]. By narrowing the eye's opening and reducing unfocused light rays, squinting can effectively increase visual focus and acuity, providing a plausible physiological basis for why this particular facial expression is so strongly linked to changes in perceived readability. Likewise, the facial regions of the cheeks, mouth, and nose were frequently represented among the top discriminators – likely because they mimic eye squinting behavior. The exploratory data analysis also revealed a significant trend within the facial regions of the mouth, cheeks, and nose (cf. Fig. 5(b)). The results suggest that the hypothesis of distinct facial patterns between effortless reading and reading difficulties cannot be readily dismissed. Moreover, the AUC of the maximum *CheekSquint* feature was 0.82 (95% CI, 0.78–0.87), which is approximately in line with the reported AUC of 0.86 (95% CI, 0.78–0.94) by *Chen et al. (2023)* [42]. Although *Chen et al. (2023)* predicted visual impairments in young children primarily based on gaze direction and facial gestures, their and the underlying findings affirm the significant role of facial expressions in identifying readability challenges.

B. Implications

The results demonstrated largely uniform patterns during reduced readability, suggesting similar behavioral patterns occur during reduced vision. Specifically, squinting of the cheeks and associated facial movements might serve as a digital biomarker for visual health. Facial metrics were recorded from a distance of three meters, implying that tracking facial characteristics even closer to the phone – such as when interacting with it or holding it near the face while looking into the distance – might accurately detect indicators of vision loss, particularly when tracking changes over a long period. The study data showed high heterogeneity among participants, for example in the intensity of facial expressions, implying that changes in individual facial expressions need to be considered on a per-person basis rather than pursuing a one-threshold-fits-all strategy. The practical implications are that extending the approach into non-standardized settings through a long-term study is both feasible and justified. This extension would mitigate the current limitations of the study, some of which stem from the necessity of establishing a proof-of-concept before progressing to more sophisticated and time-demanding procedures. Detecting signs and progression of vision impairment using mobile devices as a long term perspective would have a significant impact on global health, considering the high prevalence and rapid increase of eye diseases.

C. Limitations

The experiments were conducted in a standardized environment to minimize external influences on the vision testing process. Nonetheless, there remains a risk that the observed behavioral changes could be attributed to non-vision-related factors, such as adaptation to the testing environment, which might alter facial expressions. Additionally, the head was not fixed during testing, allowing for minor movements in distance and angle relative to the smartphone, which could impact the accuracy of the results. While samples with detected open mouths were excluded, interactions between speech dynamics and other facial landmarks, like the cheeks, might still introduce bias into the analysis. The generalizability of facial expressions in visually impaired people in non-standardized environments is also still limited.

A key limitation of this study is the relatively small and homogeneous sample of 37 participants, which restricts both the statistical power and the generalizability of the findings. With a limited number of participants, subtle differences or variations in facial reactions may not have been detectable, increasing the risk of type II errors. The absence of participants with more diverse forms of visual impairment beyond mild to mid-severe myopia and hyperopia, as well as the lack of individuals from a broader range of ages and ethnicities, further constrains the applicability

of these results. For instance, older adults or those with age-related ocular conditions may exhibit distinct behavioral patterns not captured in this cohort. Similarly, a more ethnically diverse population could present a wider array of facial expressions and cultural variations in nonverbal communication. Although steps were taken to standardize data and consider individual thresholds, the extent to which these findings can be generalized to different populations and clinical contexts remains an open question. Future studies should incorporate larger, more diverse cohorts to enhance the robustness and significance of the results.

Regarding the test distance, conducting the examination at more than three meters – according to DIN Norm a minimum distance of four meter is required [48] – presents challenges due to the limited screen size of smartphones, which restricts the display of large letters. Conversely, when testing too close to the smartphone screen the far point of people with low myopia might lie beyond the test distance. Their visual impairment will not be noticable, no matter how small the letters get. Consequently, the current implementation is most effective within a range of two to three meters.

Compared to existing remote vision testing tools, the application currently falls short in terms of usability. While it includes an onboarding process and guides the user through the test, there is room for improvement, particularly in providing visual guidance and audio instructions. These improvements would allow the test to be carried out independently and without the help of a second person. In addition, the results of facial recognition are not processed or displayed directly on the device, but the data must be transferred to a computer for further analysis. This requirement restricts the use of the application in its current form to the research field and does not make it accessible for wider personal use.

D. Future Work

The application's scope can be extended by incorporating additional data, such as voice patterns and posture, enabling analyses akin to those performed with facial features. A potential hypothesis is that diminished readability correlates with changes in voice confidence. Integrating a remote control feature would allow optometrists to administer tests via telehealth platforms. Additionally, collaborative learning of blendshape thresholds using secure aggregation protocols could enhance the identification of crucial vision-related factors, all while preserving user privacy. Cross-platform recording and analysis of facial features could be a valuable step into the future, allowing the application to be used across a wide range of mobile devices as well as desktop and laptop computers.

To validate and expand upon the findings of this study in a practical context, the authors plan to integrate facial feature tracking into common, non-standardized everyday environments, such as web browsers and e-book readers. The proposed roadmap involves first consolidating the current facial landmark tracking and external data analysis components – currently performed post-hoc on a central computer – into a standalone mobile library that can be natively deployed on popular operating systems. In addition, the authors will integrate other sensor data (e.g., device orientation, camera-to-user distance, and ambient light intensity) to automatically adjust for environmental variations that may affect vision behavior. Once the library is fully integrated, for example into an open-source web browser platform, frequent on-device data sampling can occur during the user's interaction.

The authors plan to implement a user feedback mechanism via notifications. Periodically, users may be asked to provide subjective assessments of their perceived vision quality, as well as the results of any recent professional eye examinations. These user inputs can serve as *soft labels* that, combined with the continuous facial expression data, will help correlate observed changes in facial behavior with variations in visual acuity over time. Additionally, research into user experience and communication strategies must be conducted to ensure users are sensitively informed about changes in their data (e.g., significant increases in squinting strength).

A clinical study is planned, in collaboration with ophthalmologists, to further evaluate the feasibility and accuracy of these methods in both inpatient and remote settings. This initiative will involve regular professional eye examinations and ongoing user feedback, strengthening the clinical evidence and applicability of the proposed solution. With a larger, more diverse participant cohort, it will be possible to advance with more sophisticated AI models such as tree-based algorithms. These advances could lay the foundation for a widely available digital solution for the unobtrusive detection of signs of visual impairment.

V. Conclusion

This study explored the use of mobile devices to conduct a standard eye examination, capturing facial features to identify patterns indicative of wrong letter reading, which may suggest vision impairment. The adaptation of the *logMAR* test to a mobile environment showed no technological constraints. While the test effectively differentiated between severe and mild impairment, it struggled to identify more subtle variations in vision. Analysis of facial features uncovered patterns related to the size of letters perceived, thus affecting reading ability. Preliminary findings suggest that gestures such as cheek squinting could predict vision impairment, though further validation in larger, longer-term studies in non-standardized settings is required. Future research should consider the inclusion of additional sensor data, cross-platform compatibility, and integrating facial feature tracking into daily-use applications. This approach aims to continuously monitor facial behaviors linked to vision issues, complemented by regular eye exams for significant evaluations. Thereby, on-device monitoring and analysis of behavioral data for detecting ongoing vision deterioration provide a privacy-preserving and continuous assessment tool that is available independent of time and location.

Declaration of AI-Assisted Technologies in the Writing Process

During the preparation of this work, the authors used AI to check the text for readability and to eliminate typos and grammatical errors. Subsequently, the authors reviewed and

edited the content as needed and take full responsibility for the content of the publication.

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